

Title of the LAB Based Project :-

INTERACT: Design and Development of an Integrated Social Media Application Using React Native and Supabase .

Abstract:- Interact is a cross-platform social media application developed using React Native and Supabase. It combines content creation, discovery, discussion, and real-time communication in a single platform. The application supports videos, short clips, images, text posts, stories, and live interactions. The system focuses on usability, scalability, and performance while providing a unified social media experience.

Problem Statement :- Existing social media platforms focus on limited content formats and require users to switch between multiple applications for different needs such as messaging, video sharing, and discussions. This results in fragmented user experience and inefficiency.

Aim of the Project :- To design and develop a unified social media application that supports multiple content formats and real-time communication using modern cross-platform technologies.

Scope of the Project :- The project focuses on building a mobile-based social media platform that integrates content sharing, messaging, and discovery features. It is scalable and can be extended with advanced AI-based recommendations and analytics in the future.

Objectives :-

- ✚ To develop a cross-platform mobile application using React Native .
- ✚ To integrate secure authentication and database services using Supabase .
- ✚ To support multiple content types such as videos, images, and text .
- ✚ To provide real-time messaging and content interaction .
- ✚ To ensure smooth performance and user-friendly interface .

Software Requirements :-

Operating System: Windows / macOS / Linux

Frontend Framework: React Native

Backend & Database: Supabase

Programming Languages: JavaScript, SQL

Development Tools: VS Code, Android Studio

Hardware Requirements :-

Processor: Intel i3 or higher

RAM: Minimum 8 GB

Storage: Minimum 10 GB free space

Mobile Device or Emulator for testing

System Architecture :- The system follows a client-server architecture where the React Native mobile application communicates with Supabase services for authentication, database operations, real-time updates, and media storage.

Modules Description :-

Authentication Module: Handles user login and registration .

Content Module: Manages posts, videos, moments, and stories .

Chat Module: Provides real-time messaging functionality .

Search Module: Enables content and user discovery .

Profile Module: Manages user profiles, history, and saved content .

Advantages :-

Unified platform for multiple social media features

Cross-platform support

Real-time communication

Scalable and secure backend

Applications :-

Social networking

Content creation and sharing

Community discussions

Real-time messaging

Future Enhancements :-

AI-based content recommendation

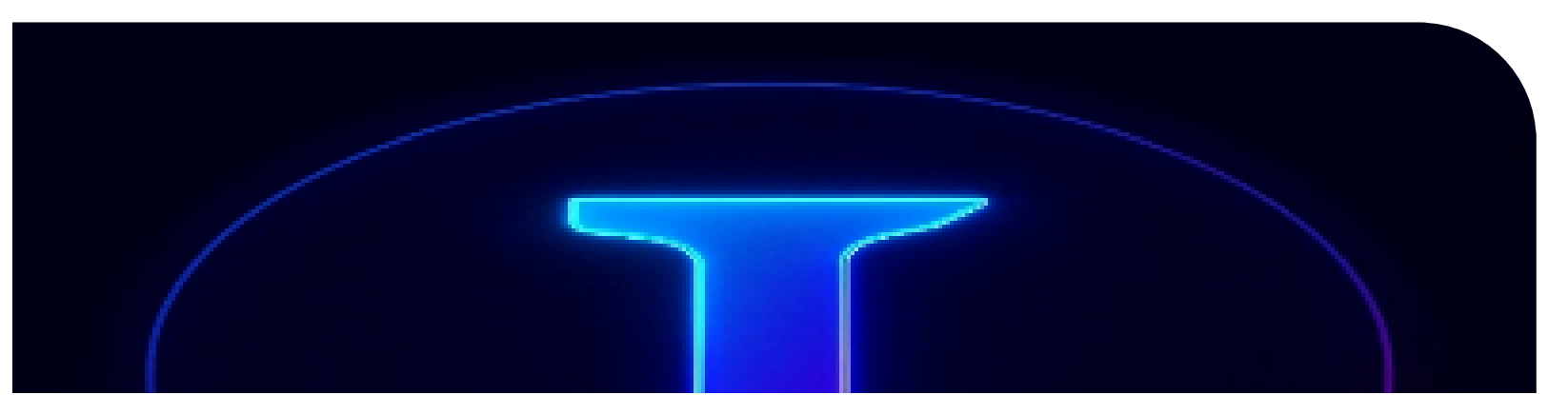
Monetization features for creators

Advanced analytics dashboard

Web version of the application



Product Requirements Document (PRD):



Attribute	Specification
Product Name	Interact
Version	1.0 (Initial Production Release)
Platform	React Native (iOS&Android)
Objective	Deliver a unified social media platform that blends content creation, discovery, discussion, and real-time interaction into a single, cohesive experience.
Target Users	Everyday social media users seeking one platform for video, photos, text, live content, and private communication.
Backend&Database	Supabase (Interact Database)
Default UI Mode	Dark Mode
Release Status	Production-Ready

Interact App :-

1. Product Goals & Success Metrics :-

1.1 Core Objectives :-

- Launch a **fully functional, stable, and scalable** cross-platform mobile application.
- Offer a smooth, intuitive experience despite a broad and diverse feature set.
- Ensure zero dead clicks. Every interaction must do something meaningful.

1.2 Key Performance Indicators (KPIs) :-

User Growth: 10,000 active users within 90 days of launch.

Retention: DAU/MAU ratio ≥ 20%.

Engagement: Average session duration ≥ 10 minutes .

Content Creation: Minimum of 5 new content items per DAU per week across all formats.

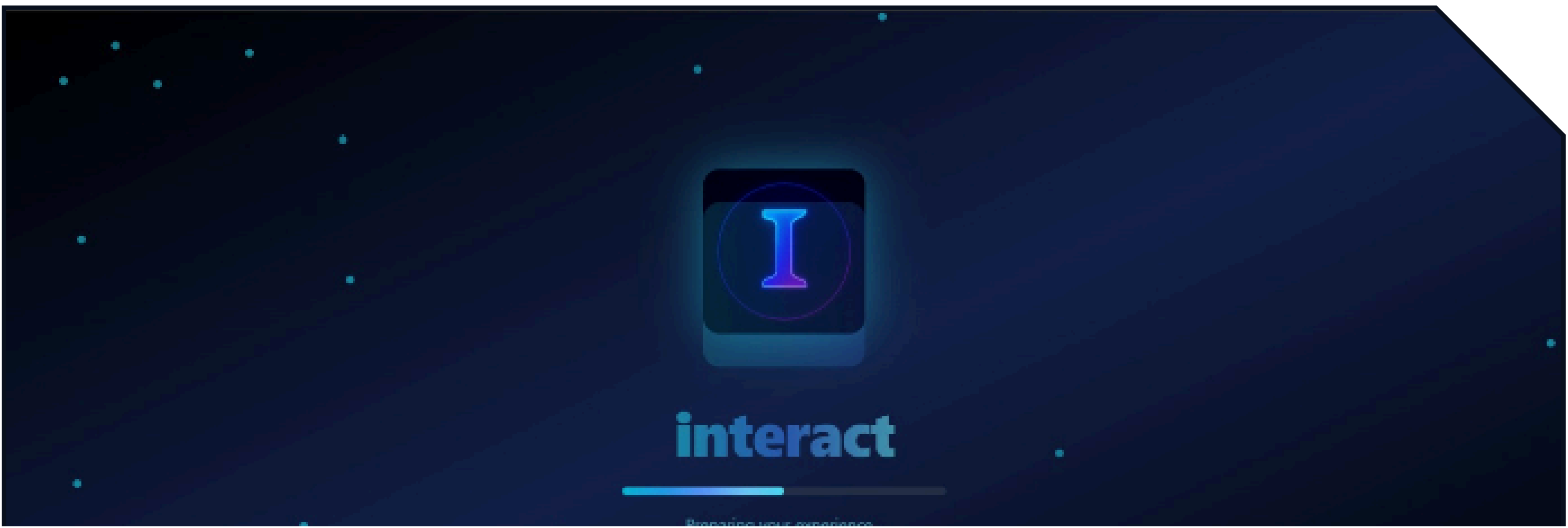
2. ⚙️ Technical & System Requirements :-

- Frontend**:- React Native for iOS and Android parity.
- Backend Services**:- Supabase for authentication, database management, storage, and real-time updates.
- Scalability**:- System architecture must support high concurrency and large-scale media uploads.
- Performance Standards**:-
 - Smooth transitions and animations.
 - Instant media playback with minimal buffering.
 - Optimized feeds for low-latency scrolling.

3. 🌐 User Experience & Navigation :-

3.1 Onboarding&Authentication :-

Feature	Requirement
Splash&LoadingScreen	Custom Interact logo with a unique animated splash experience.
Authentication	Unified Login/Signup using email-based authentication via Supabase.



3.2 Primary Navigation:- A custom-designed bottom navigation bar with six core sections:

- i. Home
- ii. Discover
- iii. Create
- iv. Moments
- v. Thoughts
- vi. Profile

Each tab must use a distinct, custom icon aligned with Interact’ s brand identity.

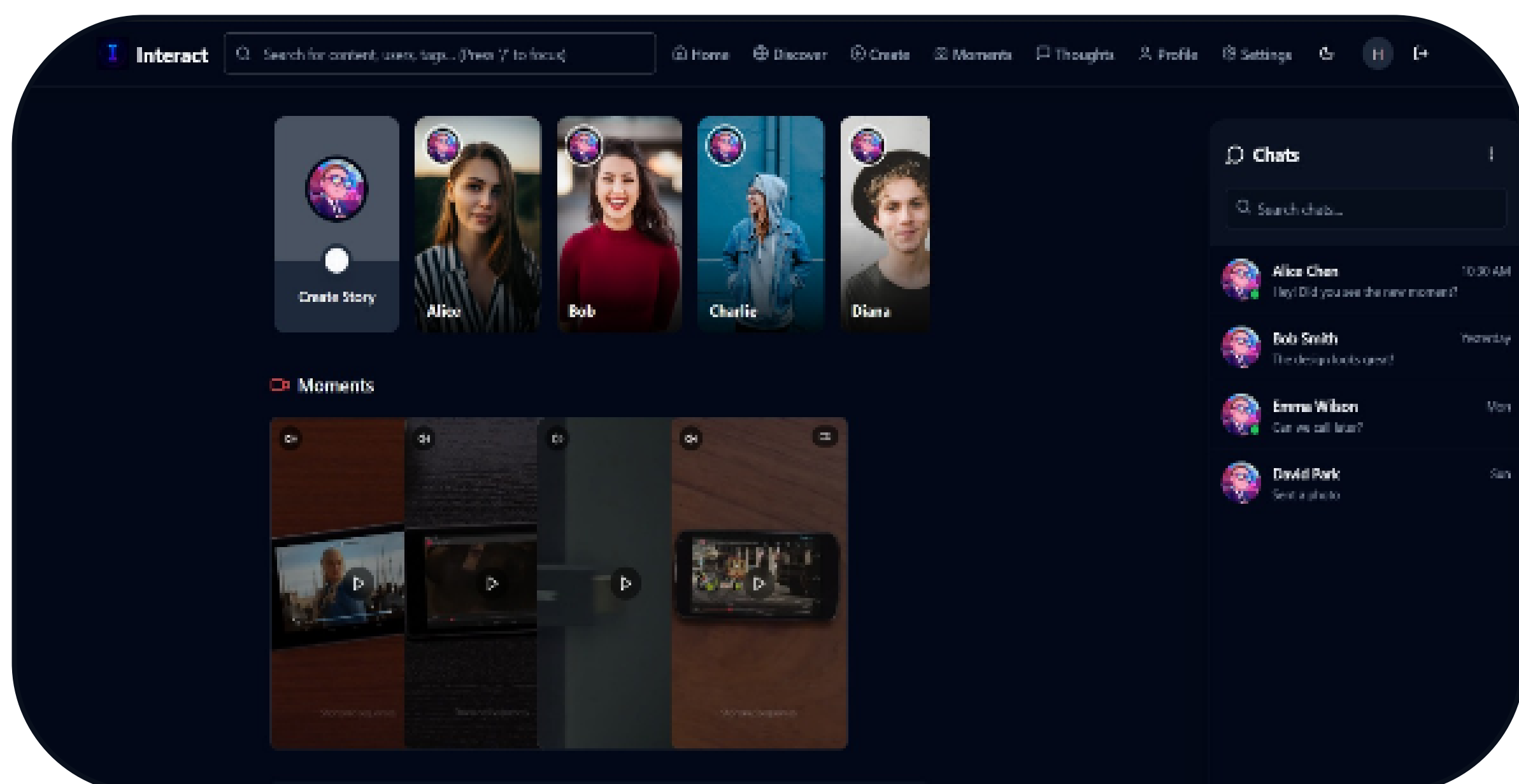




Fig 3.2 Primary Navigation Bar of Interact Application

3.3 Global Search (Top Header) :-

- a. Persistent, fixed-position search bar across all main screens.
- b. AI-powered search across users, videos, thoughts, posts, and trends.
- c. Real-time intelligent suggestions based on user behavior and trending topics.

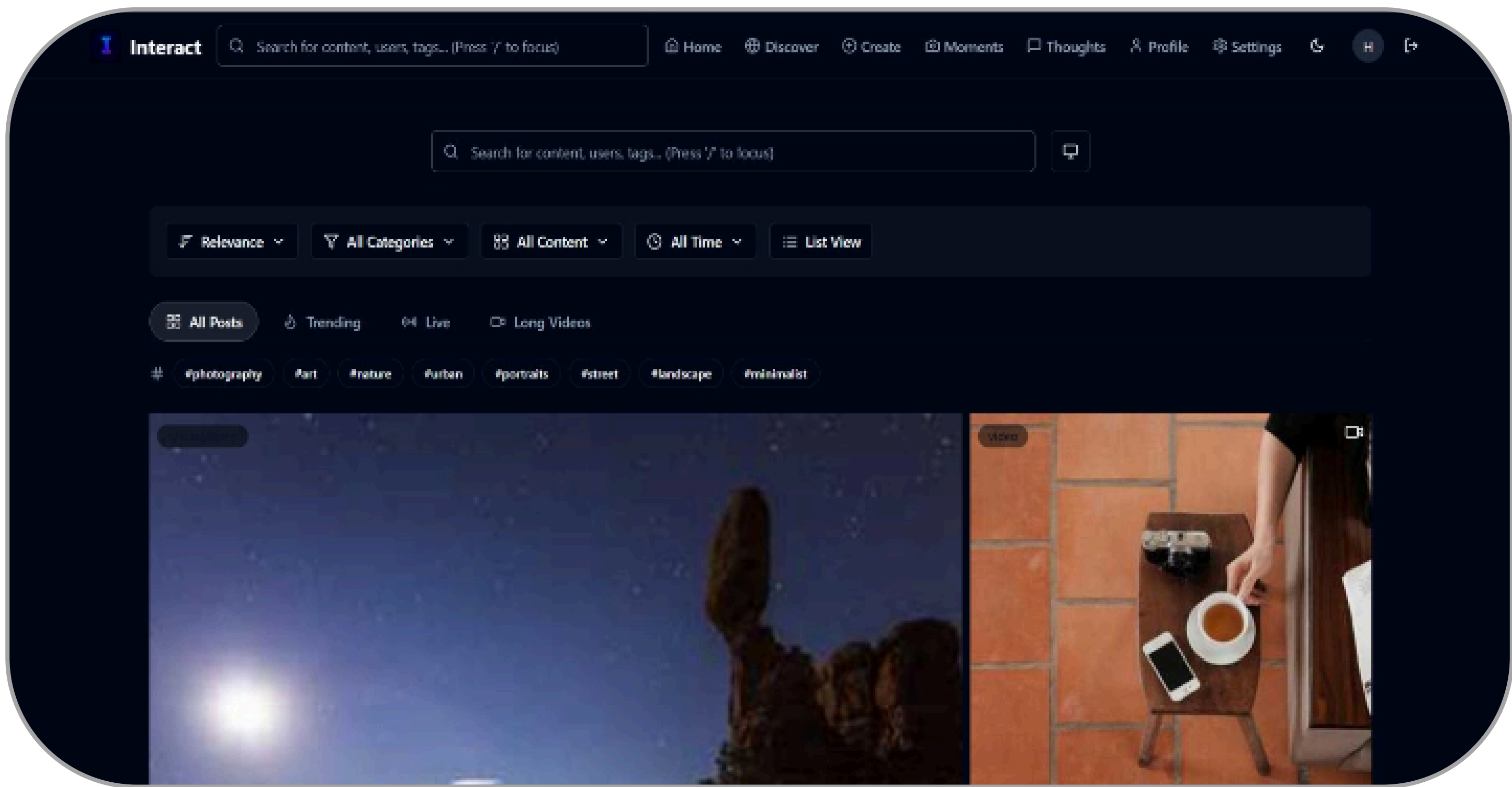


Fig 3.3 Home Screen of Interact Application

3.4 Theme Management :-

- Dark Mode enabled by default.
- Global toggle allowing users to switch between Dark and Light themes instantly.

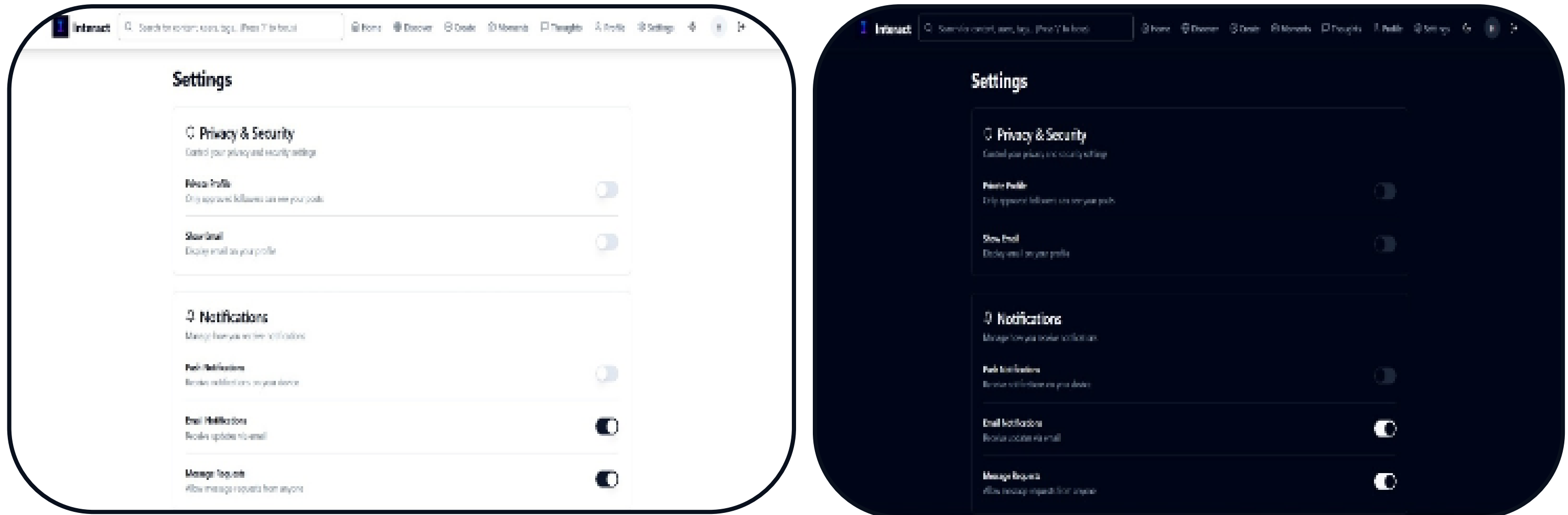


Fig 3.4 Theme and User Preference Settings Screen

4. 📱 Feature Breakdown by Page :-

4.1 Home (Main Feed) :-

Component	Description
UnifiedFeed	Mix of followed and recommended content (videos, photos, thoughts).
Story Panel	Ephemeral content carousel inspired by Stories, redesigned uniquely for Interact.
MomentsCarousel	Horizontal section highlighting short-form videos.
ChatsSidebar	Slide-out chat panel for direct messaging.

4.2 Chats Sidebar :-

Capability	Description
AccessRules	Messaging allowed between mutual followers or one-way followers.
Messaging	Real-time text chat.
Attachments	Images, files, GIFs supported.
Chat Customization	Custom themes using RGB color selection or uploaded images/GIFs.
Media Editing	Built-in basic editor for images (crop, resize, filters).

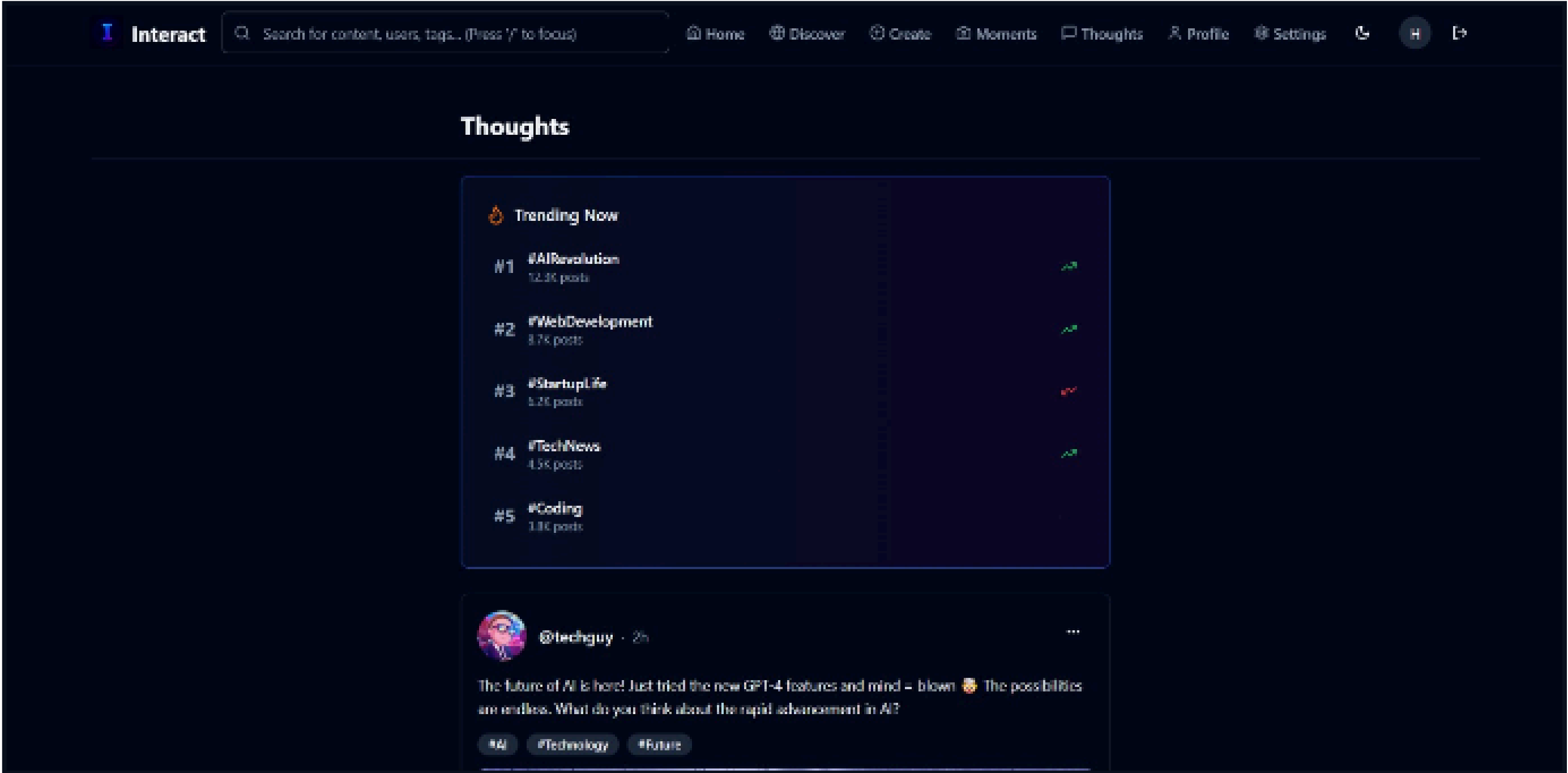


Fig:- Thoughts – Trending Discussions Screen

Section	Description
Explore Grid	Visual discovery feed with cards and grids.
Trending	Rapidly growing and high-engagement content.
Live Streams	Real-time streams with live chat functionality.
Long Videos	Dedicated long-form video layout with thumbnails, metadata, and views.

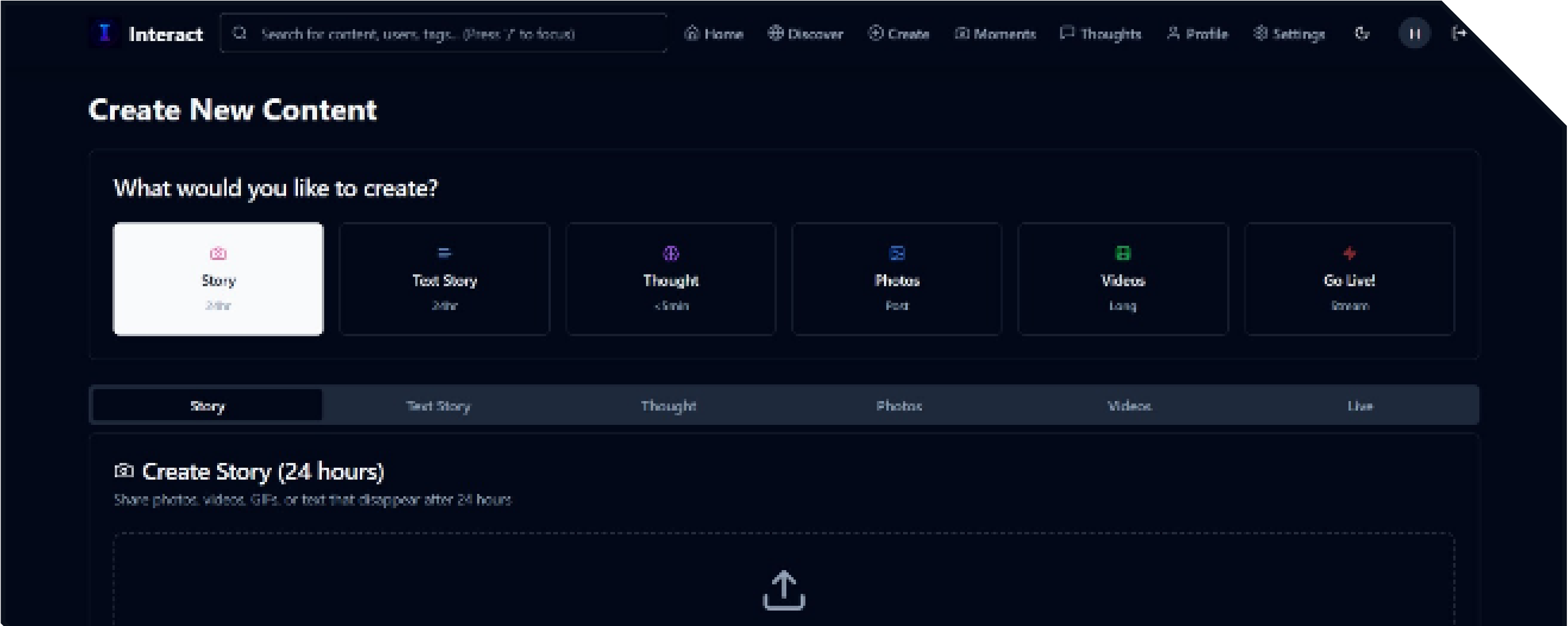
4.3 Discover (Explore) :-

4.4 Create :-

Supported content formats :-

- a. Long-form videos
- b. Short vertical videos (Moments)
- c. Image posts (single or multiple)
- d. Text-based thoughts
- e. Stories (photos or short videos, ephemeral)

Each format must have a clear, guided creation flow .



4.5 Moments :-

- a. Full-screen vertical feed for short videos only.
- b. Continuous, swipe-based consumption.
- c. Optimized for high engagement and rapid discovery.

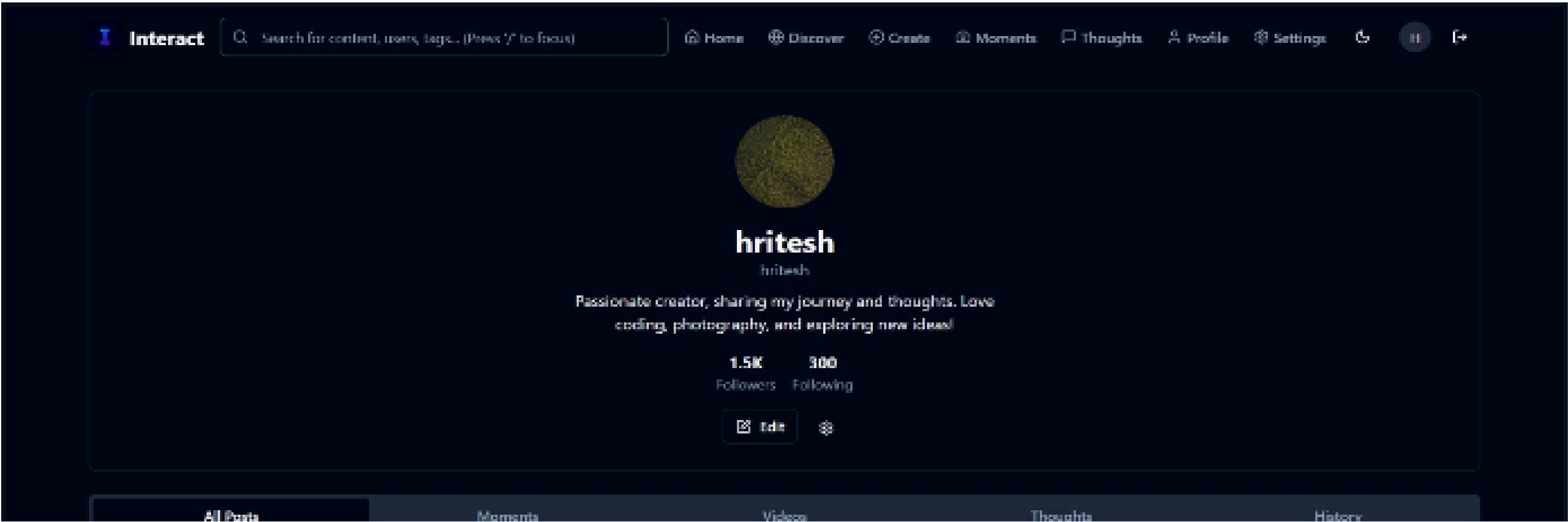
4.6 Thoughts :-

Text-first social space blending discussion and microblogging.
Features include :-

- Like, Upvote, Downvote
- Commenting and threaded replies
- Sharing across the platform

Section	Content
Content Tabs	All Posts, Thoughts, Moments, Videos, Photos
History	Complete browsing and viewing history within the app
Saved	User-saved content across all formats

4.7 Profile :-



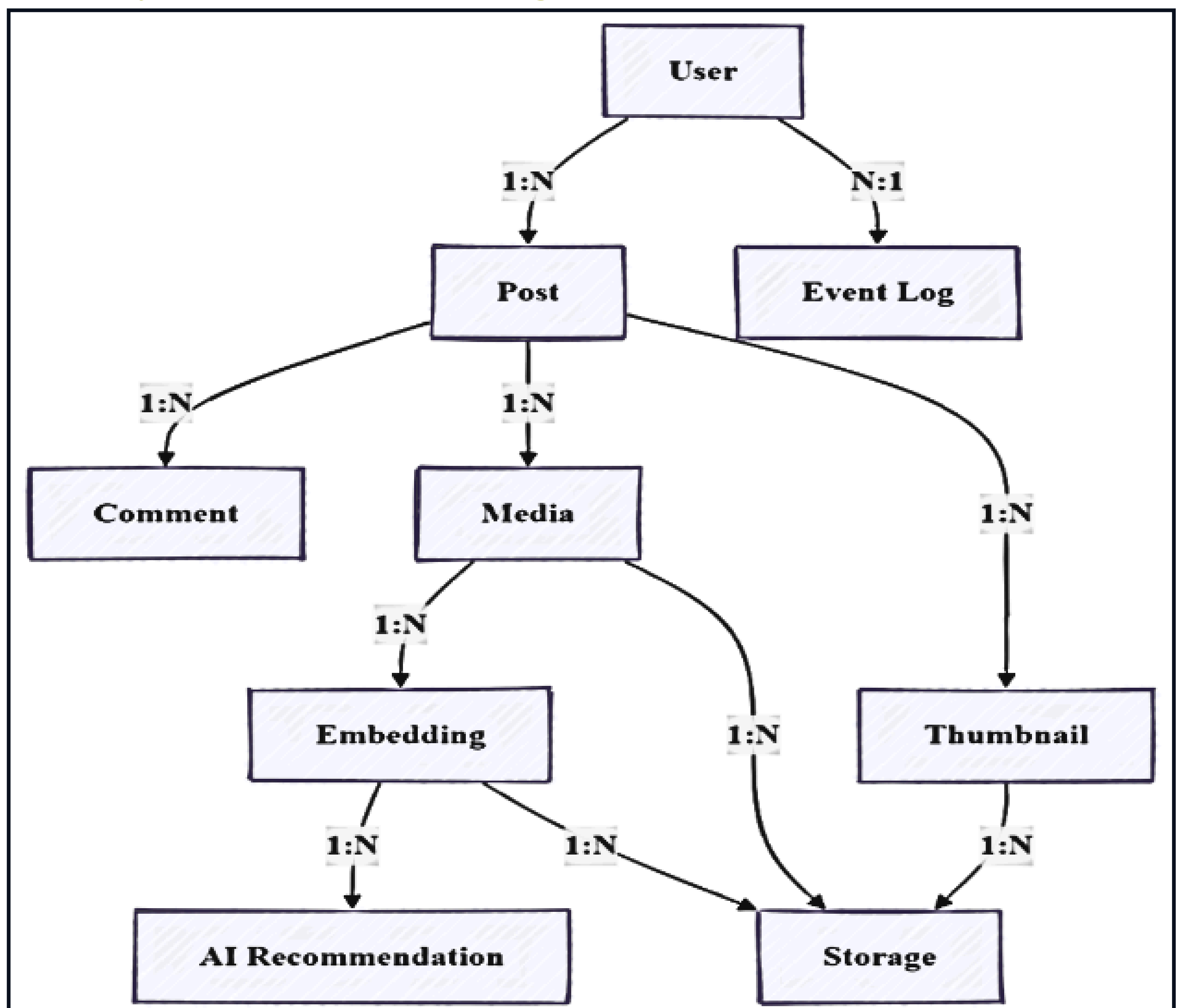
5. 🛠️ Quality Assurance & Release Readiness :-

- i. All documented features must be fully implemented.
- ii. No broken navigation paths or inactive buttons.
- iii. Responsive layouts across screen sizes and orientations.
- iv. Thorough unit, integration, and end-to-end testing.
- v. Secure authentication and encrypted data transmission (HTTPS/WSS).
- vi. Safe handling, sanitization, and storage of user-generated content.

6. 🖼️ Branding & Visual Identity :-

- i. Consistent use of the Interact logo across splash screens, headers, and icons.
- ii. Fully custom iconography for navigation and actions.
- iii. UI inspiration allowed, direct visual copying is not.
- iv. Interact must feel familiar, not derivative.

7. Entity Relationship (ER) Diagram :-



7.1 ER Diagram Explanation :- The ER diagram defines the database structure of the **Interact** application and illustrates how different entities are connected to support content creation, media handling, and AI-based recommendations.

Entities Description :-

User :- Stores user details such as name, email, password, and profile information .

Post :- Stores user-generated content including text posts, images, and videos .

Event Log :- Stores user activity records such as logins, post interactions, and system events .

Comment :- Stores comments made by users on posts .

Media :- Stores media files such as images and videos associated with posts .

Thumbnail :- Stores preview images generated for posts and media content .

Embedding :- Stores vector representations generated from media for AI processing .

AI Recommendation :- Stores recommendation results generated using embeddings .

Storage :- Represents the storage system where media files, thumbnails, and embeddings are saved .

Relationships :-

One **User** can create multiple **Posts** .

One **User** can generate multiple **Event Logs** .

One **Post** can have multiple **Comments** .

One **Post** can contain multiple **Media** files .

One **Post** can generate multiple **Thumbnails** .

One **Media** can have multiple **Embeddings** .

One **Embedding** can generate multiple **AI Recommendations** .

Media, Thumbnail, and Embedding entities are stored in **Storage** .

7.2 ER Diagram – Primary Key (PK) and Foreign Key (FK) Table :-

Entity	Primary Key	Foreign Key	Description
User	user_id	–	Stores user information
Post	post_id	user_id	Content created by user
Event Log	event_id	user_id	Stores user activity logs
Comment	comment_id	post_id, user_id	Stores comments on posts
Media	media_id	post_id	Stores images and videos

Thumbnail	thumbnail_id	post_id	Stores preview images
Embedding	embedding_id	media_id	Stores vector data
AI Recommendation	recommendation_id	embedding_id	Stores recommendation results
Storage	storage_id	—	Stores media and data files

8. System Architecture Diagram :-

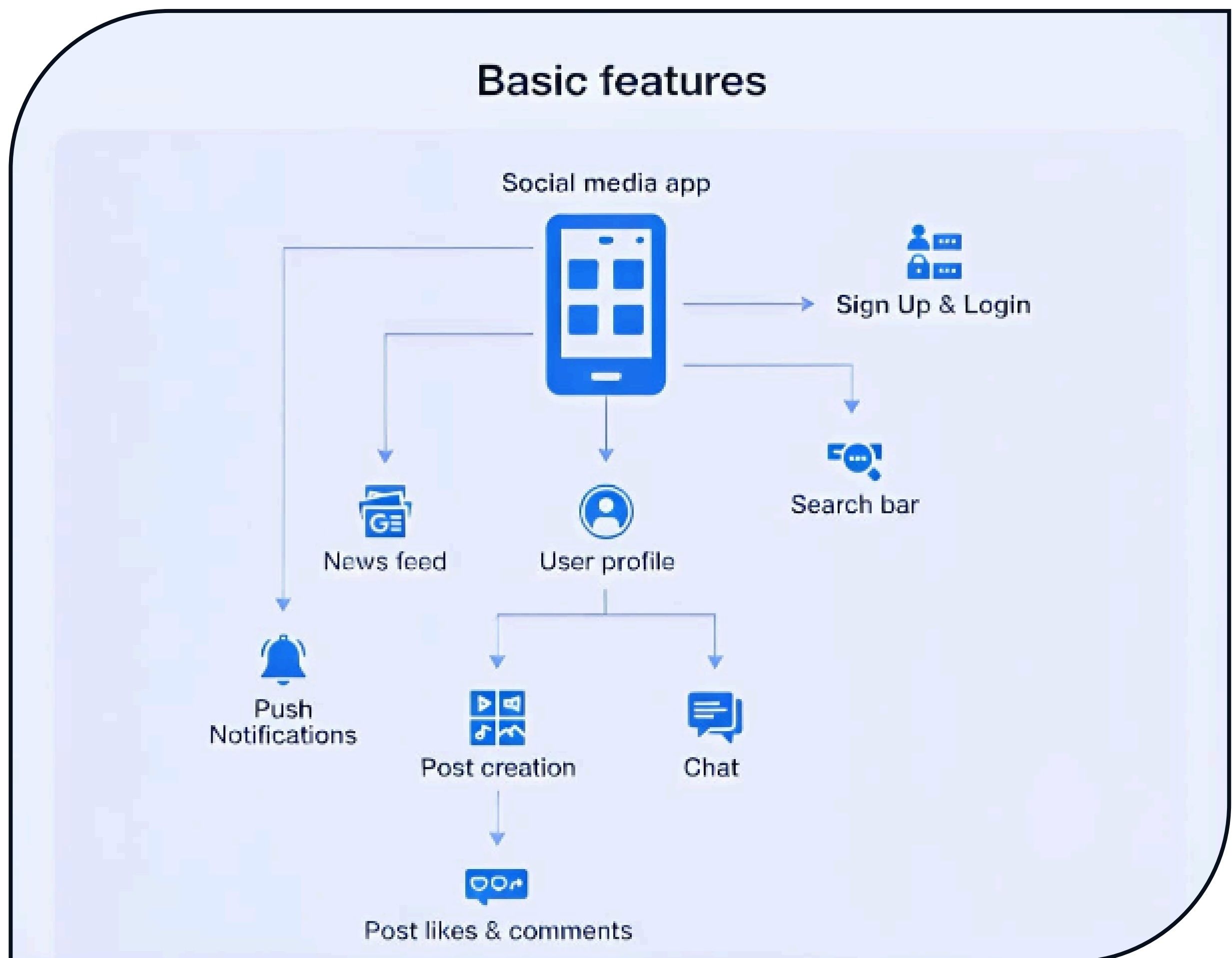


Fig:- System Architecture Diagram of Interact Application

8.1 System Architecture Explanation :- The system architecture of the **Interact** application is designed based on the entities and relationships defined in the ER model. It follows a **client– server architecture** to support content creation, media handling, storage, and AI-based recommendations.

Architecture Components (According to ER Model) :-

User Interface Layer (Client Side) :- The frontend of the application is developed using **React Native**. Users interact with the system through this layer to create posts,

upload media, view content, comment on posts, and manage their profiles. All user actions are initiated from this layer .

Application Layer :- This layer processes user requests and manages core application logic such as post creation, comment handling, media uploads, and event tracking. It acts as a bridge between the user interface and backend services.

Backend Services Layer (Supabase) :- This layer handles authentication, database operations, and real-time communication. It manages entities defined in the ER model such as :

- a. **User**
- b. **Post**
- c. **Comment**
- d. **Event Log**

All data transactions are securely handled using Supabase services .

Media Processing Layer :- When a user uploads media content, it is processed in this layer. Media files are associated with posts and further used to:

Generate **Thumbnails** for preview

Create **Embeddings** for AI-based analysis

AI Recommendation Layer :- The embedding data generated from media is used in this layer to produce **AI Recommendations**. This helps in content discovery and personalization within the application .

Storage Layer :- The **Storage** component stores all large data objects such as :

- Media files
- Thumbnails
- Embeddings

This layer ensures secure and scalable storage as defined in the ER model.

Data Flow Based on ER Model :-

User creates a **Post** through the client application

Post may include **Media** files

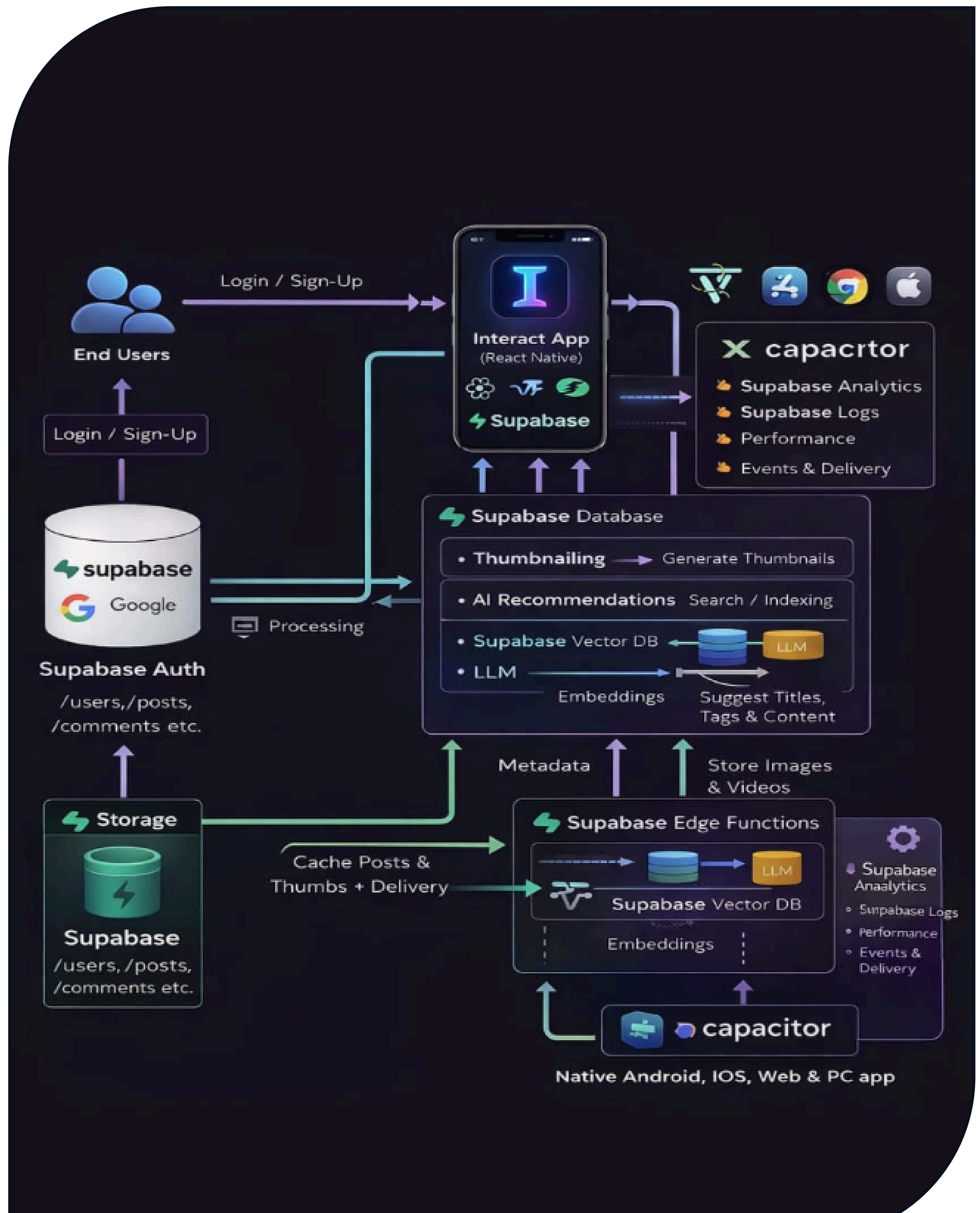
Media files are stored in **Storage**

Media generates Embeddings

Embeddings are used to generate AI Recommendations

User interactions generate **Comments** and **Event Logs**

Thumbnails are generated for media and stored in **Storage**



Advantages of the Architecture :-

- Matches the ER model structure
- Scalable and modular design
- Efficient media and storage handling
- Supports AI-based recommendations
- Secure client– server communication

9. Result:- The Interact application successfully integrates content sharing, real-time messaging, and user interaction into a single platform. The system architecture ensures scalability, security, and smooth performance .

10. Conclusion:- This LAB-based project demonstrates the successful design and development of a modern social media application using React Native and Supabase. The project fulfills all functional and technical requirements and provides a scalable solution for unified social networking .